Exp 10: Applying Performance Optimization Techniques (Indexing, Subforms, Query Tuning) to a Zoho Creator Application

1. Aim

To identify, configure, and evaluate performance optimization techniques for the **Loan Application & Pre-Screening Management** app in Zoho Creator. The exercise focuses on searchable field design, optional subform use, efficient Deluge criteria, and lean report queries for **Loan_Application_Form** and **Loan_Application_Report**.

2. Procedure

Step 1: Establish a measurable baseline

Create a non-production dataset that resembles the expected loan enquiry volume. Record the time required to open Loan_Application_Report, search an applicant, filter by Loan_Type, and retrieve callback records. Repeat each test under the same conditions and retain the median result.

Step 2: Select search and filter candidates

Review the fields used by loan officers. Candidate lookup fields are **PAN_Number** and **Contact_Number**; candidate filter fields are **Loan_Type**, **Employment_Type**, and **Preferred_Callback_Time**. Enable only the indexing or search options supported by the current Creator plan and confirmed by the report workload.

Step 3: Preserve the thirteen-field intake design

Keep the thirteen Basic Fields from the reference assignment unchanged. If the application later records repeated document checks or officer follow-ups, model those repeating rows in a separate related form or an optional subform instead of adding numbered fields such as Document_1, Document_2, and Document_3.

Step 4: Tune Deluge and report criteria

Push selective criteria into the record fetch, avoid fetching all applications before filtering, retrieve only the fields required by the workflow, and move repeated lookups outside loops. Test the optimized version with the same dataset and user role as the baseline.

Step 5: Verify privacy and correctness

Confirm that optimization does not expose PAN, address, income, or liability data to unauthorized users. Compare result counts before and after each change so that faster queries do not silently omit valid applications.

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3. Output

3.1 Search and Filter Optimization Matrix

Field	Use in loan workflow	Optimization decision	Validation check
PAN_Number	Exact applicant lookup	Candidate for exact-match search support	Same record returned before and after
Contact_Number	Applicant contact lookup	Candidate for exact-match search support	Country-code format remains valid
Loan_Type	Route and filter applications	Use selective report criteria; index only if supported and beneficial	Filter count matches source records
Employment_Type	Pre-screening filter	Use bounded categorical criteria	Salaried / Self-Employed values preserved
Preferred_Callback_Time	Callback scheduling	Fetch only the relevant time window	No eligible callback is omitted

3.2 Repeating Data: Flat Fields vs. Related Rows

Design	When it fits	Performance and integrity effect
Thirteen Basic Fields in Loan_Application_Form	Initial enquiry specified in Main_ref	One compact submission; simple report and validation
Optional Document_Checks subform	Several documents must be checked per application	Keeps repeated rows together; payload grows with row count
Related Loan_Follow_Up form	Long-running case history or many officer actions	Supports independent paging and access; requires keyed queries

Design rule: Do not introduce a subform solely for speed. Choose it when the data repeats and the expected row count remains manageable. For large or independently governed histories, use a related form with a case reference.

3.3 Benchmark Recording Template

Operation	Baseline median	Optimized median	Record count	Result count
Open Loan_Application_Report	To measure	To measure	Record	Record
Exact PAN lookup	To measure	To measure	Record	1 or policy-defined
Filter by Loan_Type	To measure	To measure	Record	Record
Callback time-window query	To measure	To measure	Record	Record

The reference files do not contain an executed benchmark or Creator performance trace. Populate this table only with measurements from the actual environment.

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3.4 Deluge Query Tuning Patterns

The following patterns show the intended structure. Field names and supported syntax must be confirmed in the deployed Creator application.

```
// Avoid: fetch everything, then filter inside a loop
for each application in Loan_Application_Form
{
    if(application.Loan_Type == selectedLoanType)
    {
        // process
    }
}

// Prefer: apply selective criteria when records are fetched
applications = Loan_Application_Form[
    Loan_Type == selectedLoanType
    && Monthly_Income > 0
];
```

```
// Avoid repeated applicant fetches inside a callback loop.
// Prefer one criteria-based fetch, then process the returned collection.
callbackCases = Loan_Application_Form[
    Preferred_Callback_Time >= windowStart
    && Preferred_Callback_Time <= windowEnd
];

for each caseRecord in callbackCases
{
    // Use only fields required for the callback task.
}
```

4. Result / Observations

1. The loan application workload has clear exact-match, categorical, and time-window query paths that can be tested independently.
2. The original thirteen-field intake form remains appropriate for one-time applicant data. Repeating document checks or follow-ups should use a subform or related form based on expected volume and governance.
3. Criteria-first Deluge queries reduce unnecessary record processing and make the loan officer report easier to scale.
4. A performance improvement is accepted only when median response time improves, result counts remain correct, and access controls continue to protect applicant data.

Actual latency, statement counts, and percentage improvement must be recorded after running the benchmark in Zoho Creator; this report does not treat illustrative values as measured results.